

Amazon Web Scraper Project

Python · BeautifulSoup4 · Requests · Pandas · smtplib · Jupyter Notebook

Step 1: Importing Libraries & Sending HTTP Request

```
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JupyterLab Python 3 (ipykernel)

[1]: # import libraries

from bs4 import BeautifulSoup
import requests
import time
import datetime

import smtplib

•[50]: URL = 'https://www.amazon.in/Nucleus-Science-T-Shirts-Tshirts-Chemistry/dp/B0B3MGHLPP/ref=sr_1_14?dib=eyJ2Ijo1M5J9.XrdL1l3ukkkIJQSaZmHLi7rKvj01X9wvy51UX

headers = {
    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/121.0.0.0 Safari/537.36",
    "Accept-Encoding": "gzip, deflate",
    "Accept": "text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8",
    "DNT": "1",
    "Connection": "close",
    "Upgrade-Insecure-Requests": "1"
}
page = requests.get(URL, headers=headers)

soup1 = BeautifulSoup(page.content, "html.parser")

soup2 = BeautifulSoup(soup1.prettify(), "html.parser")

title = soup2.find(id = 'productTitle').get_text()
|
price_element = soup2.find(class_='a-price-whole')

print(title)
print(price)
```

Step 2: Extracting & Cleaning Data, Writing to CSV

```
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[51]: price = price.strip()
title = title.strip()

print(title)
print(price)

Nucleus | Science T-Shirts | Science Quote Tshirts | Chemistry T-Shirt | Physics T-Shirt Printed Half Sleeve Round Neck Funny Hindi Text Graphic T Shir
t/Top for Men and Women
399

[35]: import datetime

today = datetime.date.today()
print(today)

2026-05-04

[37]: import csv

header = ['Title', 'Price', 'Date']
data = [title, price, today]

with open("AmazonWebScraper.csv", "w", newline = '', encoding = 'UTF8') as f :
    writer = csv.writer(f)
    writer.writerow(header)
    writer.writerow(data)

[49]: import pandas as pd

df = pd.read_csv(r'C:\Users\Irish Gyawali\AmazonWebScraper.csv')
print(df)

Title Price Date
```

Step 3: check_price() Function — Full Implementation

```
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[48]: with open('AmazonWebScraper.csv', 'a+', newline='', encoding='UTF8') as f:
      writer = csv.writer(f)
      writer.writerow(data)

[69]: def check_price():
      URL = 'https://www.amazon.in/Nucleus-Science-T-Shirts-Tshirts-Chemistry/dp/B083MGHLPP/'
      headers = {"User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/121.0.0.0 Safari/537.36", "Accept-

      page = requests.get(URL, headers=headers)
      soup1 = BeautifulSoup(page.content, "html.parser")
      soup2 = BeautifulSoup(soup1.prettify(), "html.parser")

      # This gets the element AND strips it in one line
      title_el = soup2.find(id='productTitle')
      title = title_el.get_text().strip() if title_el else "Title Not Found"

      price_el = soup2.find(class_='a-price-whole')
      price = price_el.get_text().strip() if price_el else "0"

      import datetime
      today = datetime.date.today()

      import csv
      data = [title, price, today]

      with open('AmazonWebScraper.csv', 'a+', newline='', encoding='UTF8') as f:
          writer = csv.writer(f)
          writer.writerow(data)
```

Step 4: Automated Daily Loop & Final CSV Output

```
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JupyterLab Python 3 (ipykernel)

[*]: while (True):
      check_price()
      time.sleep(86400)

[71]: import pandas as pd

      df = df = pd.read_csv(r'C:\Users\Irish Gyawali\AmazonWebScraper.csv')
      print(df)

      Title Price Date
0 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
1 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
2 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
3 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
4 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
5 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
6 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
7 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
8 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
9 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
10 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
11 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
12 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
13 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
14 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
15 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
16 Nucleus | Science T-Shirts | Science Quote Tsh... 399 2026-05-04
```

Step 5: check_price() with Email Alert Trigger

```
def check_price():
    URL = 'https://www.amazon.in/Nucleus-Science-T-Shirts-Tshirts-Chemistry/dp/B0B3MGHLPP/'
    headers = {"User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/121.0.0.0 Safari/537.36", "Accept-

    page = requests.get(URL, headers=headers)
    soup1 = BeautifulSoup(page.content, "html.parser")
    soup2 = BeautifulSoup(soup1.prettify(), "html.parser")

    # This gets the element AND strips it in one line
    title_el = soup2.find(id='productTitle')
    title = title_el.get_text().strip() if title_el else "Title Not Found"

    price_el = soup2.find(class_='a-price-whole')
    price = price_el.get_text().strip() if price_el else "0"

    import datetime
    today = datetime.date.today()

    import csv
    data = [title, price, today]

    with open('AmazonWebscraper.csv', 'a+', newline='', encoding='UTF8') as f:
        writer = csv.writer(f)
        writer.writerow(data)

    if(price < 300):
        send_mail()

def __init__(self):
```

Step 6: send_mail() — Email Alert via smtplib

```
def send_mail():
    server = smtplib.SMTP_SSL('smtp.gmail.com', 465)
    server.ehlo()
    #server.starttls()
    server.ehlo()
    server.login('utsavgyawali@gmail.com', 'xxxxxxxxxxxxx')

    subject = "The Shirt you want is below ₹300! Now is your chance to buy!"
    body = "Utsav, This is the moment we have been waiting for. Now is your chance to pick up the shirt of your dreams. Don't mess it up! Link here:'htt

    msg = f"Subject: {subject}\n\n{body}"

    server.sendmail(
        'AlexTheAnalyst95@gmail.com',
        msg
    )
```